PORTFOLIO PROJECT 4
DATA SCIENTIST : SEIHA VAT

Customer Retention Risk Analysis

Olist Brazilian E-Commerce · Churn Prediction Pipeline

74,899	97.82%	9 tables
Eligible customers	Base churn rate	Joined & engineered

Why Churn Prediction Matters

Olist is a marketplace aggregator

Sellers list products; Olist handles logistics and payments. Acquiring a seller or customer has a cost — retention is cheaper than acquisition.

90-day repurchase window

Industry-standard for e-commerce; beyond 90 days a customer is structurally unlikely to return without intervention.

Cost asymmetry

Missing a churner (FN) costs more than a false alarm (FP). Breakeven cost ratio = 0.61: if recovering one customer is worth >0.61× the cost of an offer, the model adds value.

R$ 13,664

Max single order value

2,801

Customers with >1 order

0.61

Model breakeven ratio

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The Data Challenge

9 tables · ~1.4M rows · Complex joins · Identity problem

9 Tables, One Customer View

customer_id ≠ customer_unique_id

Every order creates a new customer_id; only customer_unique_id tracks the true person

Payments: multi-row per order

Must aggregate before joining — voucher + credit card splits inflate row counts

Geolocation: ~1M rows

Deduplicated to 19,015 zip→state pairs before joining

The customer_id Trap

orders.customer_id → order-scoped (new ID per order)
customers.customer_id → join key only
customer_unique_id → TRUE customer identity

A customer with 5 orders has:
· 5 different customer_ids

· 1 customer_unique_id

99,441 Total orders

93,358 Unique customers (after dedup)

~6,000+ Duplicate customer_ids avoided

Using customer_id for groupby would produce silently wrong retention rates — the most common wrangling error on this dataset.

Engineering a Clean Master Table

1. Filter delivered orders — 99,441 → 96,478 rows (2,963 non-delivered excluded)
2. Aggregate payments — multi-row → one row per order: total_payment_value, max_installments, used_voucher, primary_payment_type
3. Deduplicate geolocation — ~1M rows → 19,015 zip→state pairs via modal state per zip
4. Aggregate reviews — keep latest review per order; add has_review binary flag
5. Aggregate items — item_count, total_freight, avg_item_price, n_unique_sellers per order
6. Join all tables — left joins from orders anchor outward; zero row inflation across 7 joins
7. Collapse to customer level — 96,478 orders → 93,358 unique customers
8. Apply cutoff exclusion — 93,358 → 74,899 eligible (18,459 excluded: last order after 2018-05-31)

Join step	Before	After
Delivered filter	99,441	96,478
All 7 left joins	96,478	96,478
Collapse to customer	96,478	93,358
Cutoff exclusion	93,358	74,899

02

What the Data Reveals

Before modelling, what does the dataset actually tell us?

Defining & Constructing the Churn Label

Dataset end	2018-08-29 — max delivered order date
Cutoff date	2018-05-31 — dataset_end – 90 days
Eligible	last_order_date ≤ cutoff_date — customer had time to return
churn = 1	No subsequent delivered order within 90 days of last order
churn = 0	Any consecutive order gap ≤ 90 days (consecutive-gap method)

74,899
Eligible customers

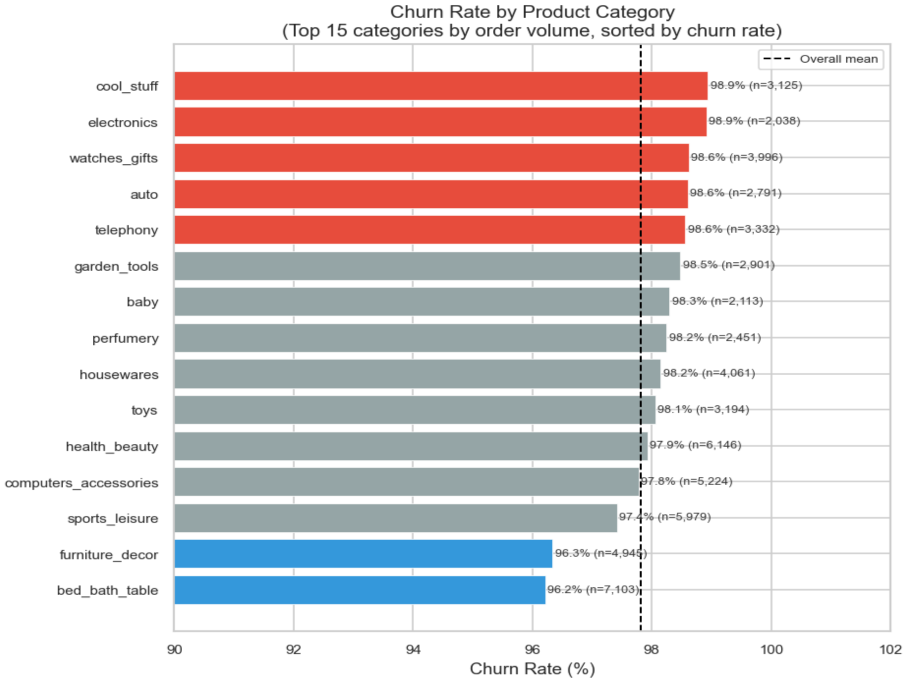
97.82%
Churn rate

2,801
Customers with >1 order

90 days
Repurchase window

Why 90 days? Shorter windows (60d) over-label seasonal buyers as churned; longer windows (120d) shrink the eligible pool too aggressively. 90 days maximises label reliability given the 25-month dataset span.

Churn Varies — But Narrowly



Product category is widest driver

bed_bath_table (96.2%) vs cool_stuff (98.9%) — 2.73pp gap, the largest single categorical spread found.

Voucher users churn less

95.65% vs 97.86% — counter-intuitive; voucher users are more engaged shoppers.

Item count is protective

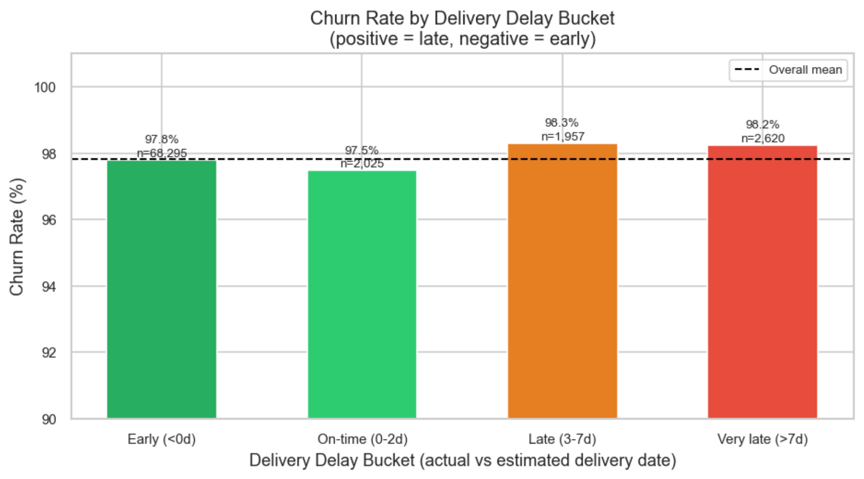
Single-item orders: 97.92% churn; 3+ item orders: ~96%.

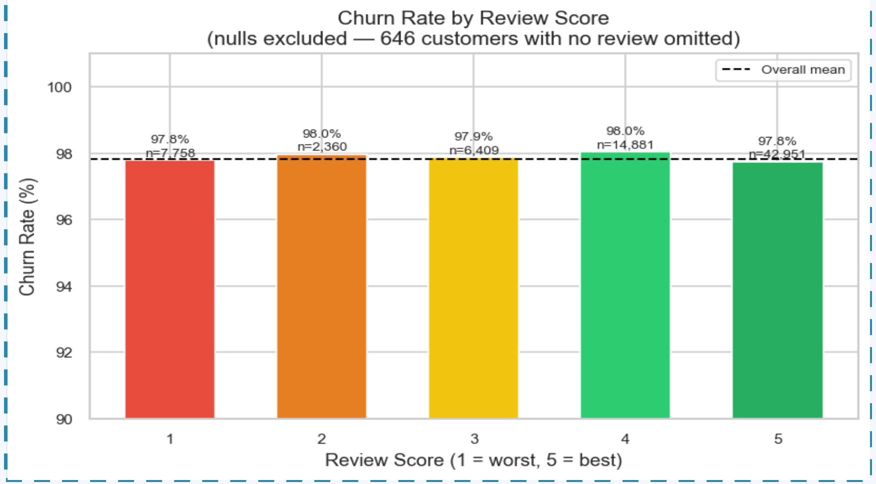
Geography matters

State churn rates range from 96.8% (MT) to 100% (RR).

The range across all variables is ≤3pp — the dataset has a structural churn ceiling near 97%. This is real behaviour, not a data quality issue.

Delivery Experience: A Surprising Pattern





DELIVERY

91.2% of orders arrived early. On-time (0–2d) customers churn least (97.48%); early arrivals churn slightly more (97.80%). The relationship is non-monotonic — delivery delay alone is not a reliable churn predictor.

REVIEW SCORE

Review score range is only 0.29pp (97.75%–98.04%). No monotonic relationship. $r_{pb} = -0.0024$, $p = 0.51$ — statistically indistinguishable from noise.

03

Feature Engineering

27 features across 7 groups — from raw tables to model-ready signals

From 28 Raw Columns to 27 Engineered Features

Temporal

n = 3 features

recency_days, is_weekend_purchase, approval_delay_days

Delivery Experience

n = 5 features

delivery_delay_days, freight_to_price_ratio, is_cross_state_order

Order Complexity

n = 7 features

item_count, max_installments, used_voucher, multi_payment_method

Monetary (log-transformed)

n = 4 features

log_avg_item_price, log_total_payment_value

Review

n = 4 features

review_score, has_negative_review, has_positive_review

Product

n = 2 features

category_churn_rate (target encoded)

Geographic

n = 2 features

state_churn_rate, seller_state_churn_rate

Monetary features log1p-transformed (3.78–4.84% extreme outliers). Target encoding for category and state — ⚠ recomputed on training set only in NB05 to prevent label leakage.

04

Modelling

Time-based split · 3 candidate models · Optuna tuning · SHAP explainability

Methodology: Why Each Decision Was Made

Time-Based Split (not random)

Train < 2018-04-01 | Test ≥ 2018-04-01

Random splitting would leak future purchase behaviour into training. Time-based split respects the temporal nature of churn.

Train: 61,696 customers · Test: 13,203 customers

class_weight='balanced'

SMOTE tested but discarded — synthetic minority oversampling inflated metrics without improving real-world discrimination.

Balanced class weights applied to all models.

Churn : Retained = 53 : 1 in training data

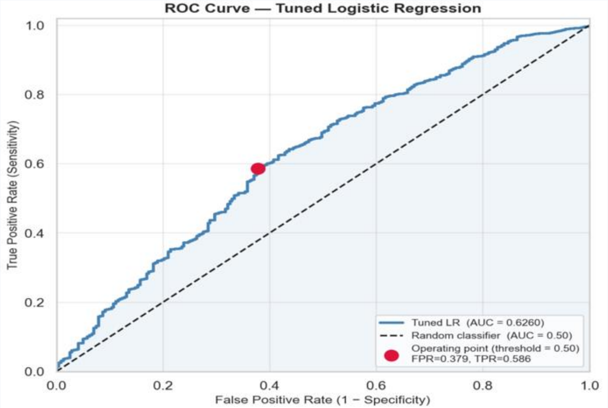
Three Candidate Models

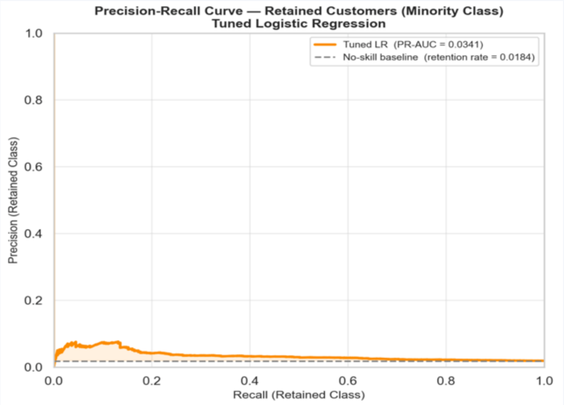
· Logistic Regression — interpretable baseline
· Random Forest — non-linear ensemble
· LightGBM — gradient boosting (primary candidate)

Tuned winning model with Optuna (50 trials, ROC-AUC objective)

Model Comparison — The Honest Results

Model	ROC-AUC	PR-AUC	Recall	Precision
Logistic Regression	0.6267	0.0343	0.617	0.027
Random Forest	0.5837	0.0268	0.210	0.027
LightGBM	0.5990	0.0310	0.284	0.032
Tuned LR (winner) ★	**0.6260**	**0.0341**	**0.621**	**0.027**





Logistic Regression wins: higher ROC-AUC and recall than RF and LightGBM despite being the simplest model. Signal in this dataset is largely linear.

Why 0.626? The Dataset Ceiling Explained

The Mathematics

- Max feature-target correlation: r = 0.07 (item_count)
- With r_max = 0.07, theoretical AUC ceiling ≈ 0.65
- 243 retained customers in test set = extremely sparse minority class
- This is not a model failure — it is a fundamental property of the data generating process

The Business Reality

- 97.82% of Olist customers buy exactly once
- Typical for general merchandise marketplaces (not subscription)
- Customers discover Olist via a specific product, buy it, and may never have a reason to return
- Repeat purchase intent is not encoded in a single transaction record

What Would Actually Help

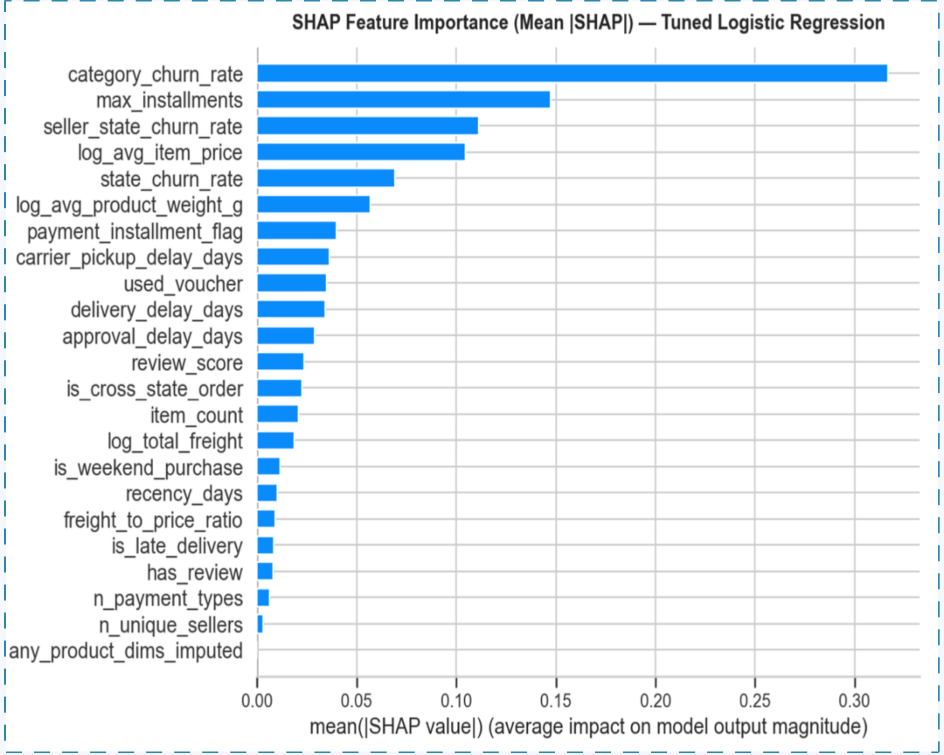
- Browsing / session history
- App engagement signals
- Return / refund history
- Multi-year purchase data
- Category affinity from search behaviour
- A richer feature set, not a better algorithm

05

What the Model Tells Us

SHAP analysis · Risk segmentation · Actionable signals

The 5 Levers of Churn Risk



0.317 — **Product category**

What was bought matters most — lifestyle/home goods retain better

0.147 — **Payment instalments**

More instalments = less churn. Offering instalment options is actionable

0.111 — **Seller geography**

Sellers in certain states have structurally higher-churn customer bases

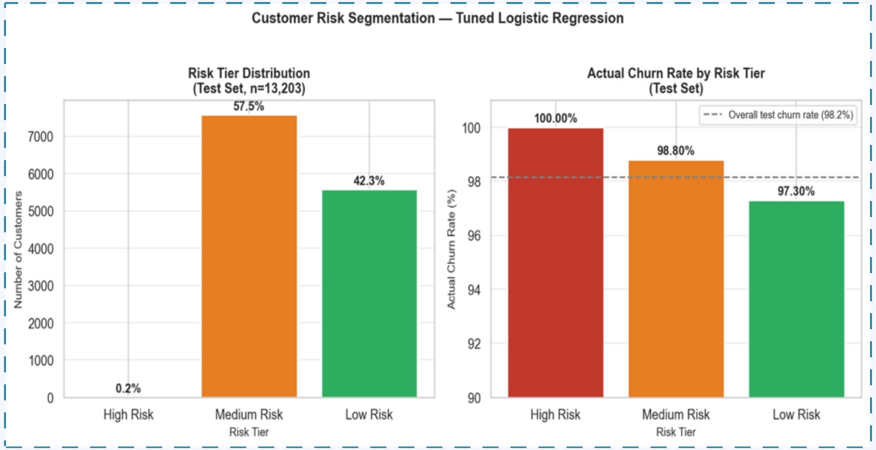
0.105 — **Item price**

Higher-priced one-off purchases signal discretionary, non-repeat buying

0.069 — **Customer state**

Regional retention patterns exist across Brazil's 27 states

Risk Segmentation: What the Model Can Do



Tier	n	Churn rate	Avg instal.	Action
High Risk (≥0.80)	24	100.00%	2.54	Low-cost re-engagement only
Medium Risk (0.50–0.80)	7,591	98.80%	2.38	Rank by score; targeted vouchers
Low Risk (<0.50)	5,588	97.30%	3.40	Bottom decile = premium outreach

Key insight: No threshold achieves precision above the 1.84% retention baseline at scale. Use churn_proba as a continuous rank, not a binary classifier, for campaign prioritisation.

Segmentation insight — voucher users in Low Risk tier show 5.3% voucher usage vs 0% in High Risk, suggesting engaged shoppers self-select into lower-risk behaviour.

What This Project Demonstrates

End-to-End Pipeline Rigour

Joined 9 tables across 1.4M+ rows; resolved customer identity; engineered 27 features with documented leakage controls; time-based train/test split.

Honest Evaluation

Model did not hit AUC ≥ 0.75. The evaluation explains why — signal ceiling, class imbalance, dataset structure. AUC = 0.626 with a clear justification is more credible than 0.91 without one.

SHAP Explainability

LinearExplainer applied to 13,203 customers. Top driver: product category (mean |SHAP| = 0.317). Individual waterfall analysis. Every finding grounded in EDA numbers.

Business Translation

Breakeven ratio 0.61 — model adds value even at modest AUC. Risk tier profiles with specific intervention recommendations. Limitations and next steps documented.

Next: Survival analysis for time-to-churn · NLP on review text · Seller-side churn analysis

https://github.com/vatsun07-web